MAT223 Chapter 8: change of basis on one page

Clips C18 to C32 on the course teaching page

Running example. $\vec{b}_1 = \begin{bmatrix}2\\1\end{bmatrix}$, $\vec{b}_2 = \begin{bmatrix}-1\\1\end{bmatrix}$, $\vec{x} = \begin{bmatrix}4\\5\end{bmatrix} = 3\vec{b}_1 + 2\vec{b}_2$, so $[\vec{x}]_{\mathcal{E}} = (4,5)$ and $[\vec{x}]_{\mathcal{B}} = (3,2)$. A second basis: $\vec{c}_1 = (1,1)$, $\vec{c}_2 = (0,1)$, with $[\vec{x}]_{\mathcal{C}} = (4,1)$.

1. What the symbols mean

- $\vec{x}$ is the arrow. It never changes.
- $[\vec{x}]_{\mathcal{B}}$ is its *address* in $\mathcal{B}$: the weights in $\vec{x} = x_1\vec{b}_1 + \cdots + x_n\vec{b}_n$.
- $M_{\mathcal{C}\leftarrow\mathcal{B}}$ is a *translator*:

$$M_{\mathcal{C}\leftarrow\mathcal{B}}\,[\vec{x}]_{\mathcal{B}} = [\vec{x}]_{\mathcal{C}}.$$

$\mathcal{B}$-coordinates go in on the right, $\mathcal{C}$-coordinates come out on the left. Read it right to left. The inner letters must match, like dominoes.

2. The column rule (every case at once)

Column j of $M_{\mathcal{C}\leftarrow\mathcal{B}}$ is $[\vec{b}_j]_{\mathcal{C}}$: the j-th input vector, written in the output basis.

Matrix	column j	running example
$M_{\mathcal{E}\leftarrow\mathcal{B}}$	$\vec{b}_j$ in standard coordinates	$\begin{bmatrix}2 & -1\\1 & 1\end{bmatrix}$
$M_{\mathcal{B}\leftarrow\mathcal{E}}$	$\vec{e}_j$ in $\mathcal{B}$-coordinates	$\frac{1}{3}\begin{bmatrix}1 & 1\\-1 & 2\end{bmatrix}$
$M_{\mathcal{C}\leftarrow\mathcal{B}}$	$\vec{b}_j$ in $\mathcal{C}$-coordinates	$\begin{bmatrix}2 & -1\\-1 & 2\end{bmatrix}$

Why, for the first column: multiplying by $\vec{e}_1$ picks out the first column, and $[\vec{b}_1]_{\mathcal{B}} = \vec{e}_1$. So

$$\underbrace{M_{\mathcal{C}\leftarrow\mathcal{B}}\,\vec{e}_1}_{\text{first column}} = M_{\mathcal{C}\leftarrow\mathcal{B}}\,[\vec{b}_1]_{\mathcal{B}} = [\vec{b}_1]_{\mathcal{C}}.$$

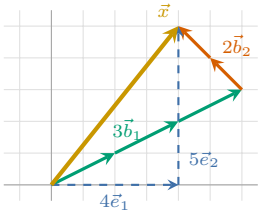
The same works for every column.

3. Why the $\vec{b}$'s are the columns of $M_{\mathcal{E}\leftarrow\mathcal{B}}$ (clip C27)

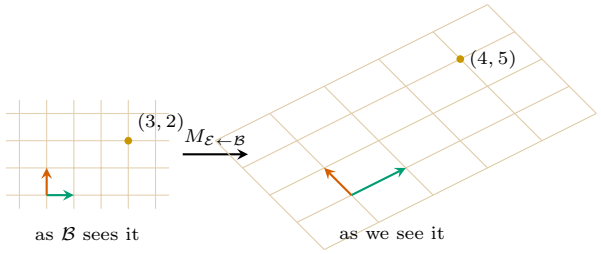
A combination of the $\vec{b}$'s, unpacked into $\vec{e}$'s, gives the standard coordinates:

$$\begin{aligned}3\vec{b}_1 + 2\vec{b}_2 &= 3(2\vec{e}_1 + \vec{e}_2) + 2(-\vec{e}_1 + \vec{e}_2)\\ &= (3\cdot 2 + 2\cdot(-1))\,\vec{e}_1 + (3\cdot 1 + 2\cdot 1)\,\vec{e}_2 = 4\vec{e}_1 + 5\vec{e}_2.\end{aligned}$$

That is $\begin{bmatrix}2 & -1\\1 & 1\end{bmatrix}\begin{bmatrix}3\\2\end{bmatrix} = \begin{bmatrix}4\\5\end{bmatrix}$. Row 1 collects the $\vec{e}_1$-parts, row 2 the $\vec{e}_2$-parts.



4. Two readings of the same matrix (clips C23, C24)



- *As a map* it sends $\vec{e}_j \mapsto \vec{b}_j$: it moves the standard grid onto the $\mathcal{B}$-grid.
- *As a translator* it sends $[\vec{x}]_{\mathcal{B}} \mapsto [\vec{x}]_{\mathcal{E}}$.

It moves grids $\mathcal{E} \to \mathcal{B}$, and that is exactly why it translates coordinates $\mathcal{B} \to \mathcal{E}$.

5. Going back, and chaining

$$M_{\mathcal{B}\leftarrow\mathcal{C}} = M_{\mathcal{C}\leftarrow\mathcal{B}}^{-1}, \qquad M_{\mathcal{C}\leftarrow\mathcal{B}}\,M_{\mathcal{B}\leftarrow\mathcal{D}} = M_{\mathcal{C}\leftarrow\mathcal{D}} \quad \text{(Exercise P8.4)}.$$

In $\mathbb{R}^n$ you may also go through $\mathcal{E}$:

$$M_{\mathcal{C}\leftarrow\mathcal{B}} = M_{\mathcal{C}\leftarrow\mathcal{E}}\,M_{\mathcal{E}\leftarrow\mathcal{B}} = M_{\mathcal{E}\leftarrow\mathcal{C}}^{-1}\,M_{\mathcal{E}\leftarrow\mathcal{B}}.$$

6. How to compute

- **One basis is standard** (Thm 8.5): $M_{\mathcal{E}\leftarrow\mathcal{B}} = \begin{bmatrix}\vec{b}_1 & \cdots & \vec{b}_n\end{bmatrix}$ and $M_{\mathcal{B}\leftarrow\mathcal{E}}$ is its inverse.
- **Two nonstandard bases** (Thm 8.8): row reduce once,

$$\left[\begin{array}{ccc|ccc}\vec{c}_1 & \cdots & \vec{c}_n & \vec{b}_1 & \cdots & \vec{b}_n\end{array}\right] \sim \left[\begin{array}{c|c} I & M_{\mathcal{C}\leftarrow\mathcal{B}}\end{array}\right].$$

Running example, in one step ($R_2 - R_1$):

$$\left[\begin{array}{cc|cc}1 & 0 & 2 & -1\\1 & 1 & 1 & 1\end{array}\right] \sim \left[\begin{array}{cc|cc}1 & 0 & 2 & -1\\0 & 1 & -1 & 2\end{array}\right].$$

- **Inside a subspace** V (LA 8.5): the vectors live in $\mathbb{R}^n$, but the RREF has zero rows at the bottom. The matrix is $d \times d$ with $d = \dim V$, not $n \times n$.

7. Three tempting mistakes, and the check (clip C31)

× $M_{\mathcal{B}\leftarrow\mathcal{E}} = [\vec{b}_1\ \vec{b}_2]$. That matrix is $M_{\mathcal{E}\leftarrow\mathcal{B}}$. It sends $(4,5)$ to $(3,9)$, and $3\vec{b}_1 + 9\vec{b}_2 = (-3,12) \neq \vec{x}$.

× Writing the $\vec{b}$'s as *rows*. $\left[\begin{smallmatrix}2 & 1\\-1 & 1\end{smallmatrix}\right](3,2) = (8,-1) \neq (4,5)$.

× $M_{\mathcal{C}\leftarrow\mathcal{B}} = [\vec{b}_1\ \vec{b}_2]$ when $\mathcal{C}$ is not standard. The columns must be $[\vec{b}_j]_{\mathcal{C}}$.

Check every answer: $M\vec{e}_1$ must be the first input vector, written in the output basis. Example: $M_{\mathcal{B}\leftarrow\mathcal{E}}\vec{e}_1 = \frac{1}{3}(1,-1)$, and indeed $\frac{1}{3}\vec{b}_1 - \frac{1}{3}\vec{b}_2 = \vec{e}_1$.